National University of Computer and Emerging Sciences,
Lahore Campus

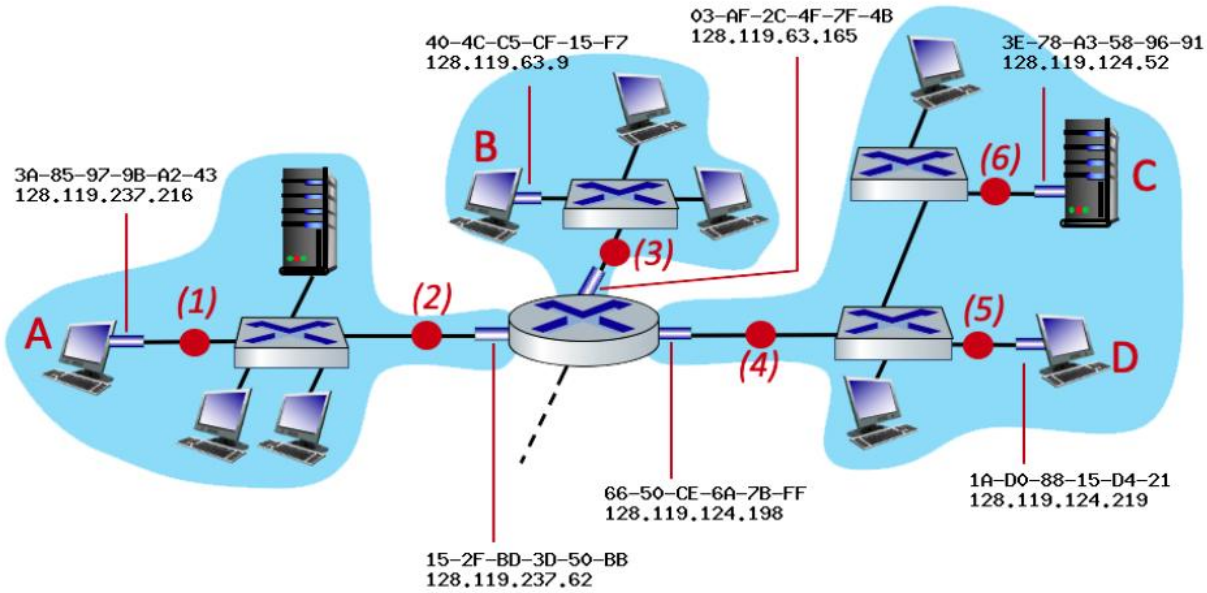
	Course Name:	**Computer Networks**	**Course Code:**	**CS 3001**
	Program:	**BS (Software Engineering)**	**Semester:**	**Spring 2026**
	Duration:	**20 minutes**	**Total Marks:**	**15**
	Paper Date:	**05-May-2026**	**Section**	**6A , 6B**
	Exam Type:	**Quiz 6 - Chapter 6**	**Page(s):**	**2**

Student Name **Roll No.** **Section:**

Q1. **Encircle the correct option:** **[1*5 = 5 Marks] [CLO 3]**

1. Host A sends a frame to Host B via Router R. Which field(s) in the layer 3 and/or layer 2 header/trailer change as it moves from Host A -> Router link -> Host B link?
   a. Source IP and Destination IP.
   b. Source MAC, Destination MAC and TTL.
   c. Source MAC and Destination MAC only.
   d. The CRC checksum only.
2. A student powers on a laptop in a coffee shop. The laptop has no static IP and the cache is empty. Which of the following is the CORRECT chronological order of the very first messages sent by the laptop?
   a. ARP Request -> DNS Query -> DHCP Discover
   b. DHCP Discover -> ARP Request -> DNS Query
   c. DHCP Discover -> DNS Query -> ARP Request
   d. SYN -> DHCP Discover -> HTTP GET
3. Consider a generator polynomial G(x) with the binary representation 101101 (6 bits). What is the maximum burst error length for which detection is guaranteed?
   a. 4 bits
   b. 5 bits
   c. 6 bits
   d. 3 bits
4. You are analyzing a generator polynomial $G(x) = x^3 + x^2 + x$ (Binary: 1110). Which of the following statements accurately describes the weaknesses of this generator?
   a. It can detect single-bit errors, but fails to detect odd-numbered errors.
   b. It can detect odd-numbered errors, but fails to detect single-bit errors.
   c. It fails to detect both single-bit errors AND fails to guarantee detection of all odd-numbered errors.
   d. It is a robust generator that detects both single-bit and odd-numbered errors effectively.
5. A 2-D parity check can:
   a. Detect a single bit error
   b. Detect & correct a single bit error
   c. Detect multiple bits errors
   d. Detect & correct multiple bits errors

Q2: Consider the figure below. The IP and MAC addresses are shown for nodes A, B, C and D, as well as for the router's interfaces. **[10 Marks] [CLO 3]**



Consider an IP datagram being sent from node B to node C. Please Answer the following:

1. What is the source mac address at point 3?

   Answer:____

2. What is the destination mac address at point 3?

   Answer:____

3. What is the source IP address at point 3?

   Answer:____

4. What is the destination IP address at point 3?

   Answer:____

5. Do the source and destination mac addresses change at point 4? Answer with yes or no. **[2]**

   Answer:____

6. What is the source mac address at point 4?

   Answer:____

7. What is the destination mac address at point 4?

   Answer:____

8. Do the source and destination mac addresses change at point 6? Answer with yes or no. **[2]**

   Answer:____